

The principle behind the pumping lemma for **Regular Languages** is that if an input string is long enough, a finite automaton processing it will have to enter some state a second time.

A corresponding principle for **CFGs** is that a sufficiently long derivation in a grammar G will have to contain a *self-embedded* variable A ; that is, the derivation looks like

$$S \Rightarrow^* vAz \Rightarrow^* vwAyz \Rightarrow^* vwxyz$$

so that the string derived from the **first occurrence** of A is wAy and the string derived from the **second occurrence** is x .

As a result,

$$S \Rightarrow^* vAz \Rightarrow^* vwAyz \Rightarrow^* vw^2Ay^2z \Rightarrow^* vw^3Ay^3z \Rightarrow^* \dots$$

As, the string x can be derived from any one of these occurrences of A , therefore, each of the strings $vxz, vwxyz, vw^2xy^2z, \dots$ can be derived from S (*hence, belongs to the language*).

The Pumping Lemma for Context-Free Languages

Suppose L is a context-free language. Then there is an integer n so that for every $u \in L$ with $|u| \geq n$, u can be written as $u = vwxyz$, for some strings v , w , x , y , and z satisfying

1. $|wy| > 0$
2. $|wxy| \leq n$
3. for every $m \geq 0$, $vw^mxy^mz \in L$

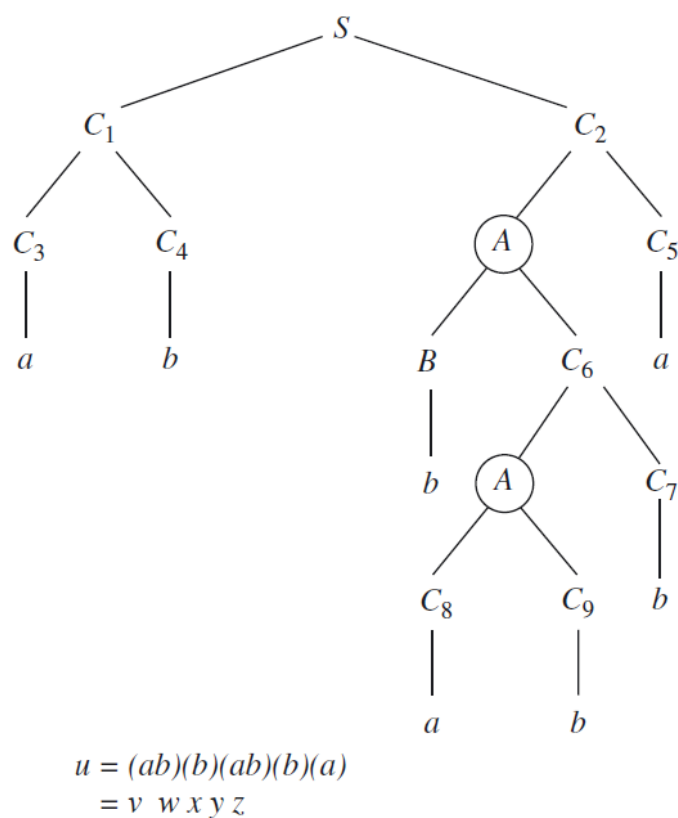
Let n be 2^{p+1} , where p is the number of **distinct variables** in the (**CNF**) grammar G , and suppose that u is a string in $L(G)$ of length at least n . Then $|u| > 2^p$, and it follows that every derivation tree for u must have height greater than p (**note that** the derivation tree of a **CNF** grammar is a **binary tree** and a binary tree of height h has no more than 2^h leaf nodes). In other words, in a derivation tree for u , there must be a path from the root to a leaf node with at least $p + 1$ **interior nodes** (nodes other than the leaf node).

Consider the portion of the longest path consisting of the leaf node and the $p+1$ nodes **just above it**. Since every interior node corresponds to a variable, and

there are only p distinct variables in the grammar, this portion of the path contains at least two occurrences of the same variable A .

Let x be the substring of u derived from the occurrence of A farthest down on the path (closest to the leaf), and let w and y be the strings of terminals such that the substring of u derived from the occurrence of A farther from the leaf is wxy . Finally, let v and z be the prefix and the suffix of u that account for the remainder of the string, so that $u = vwxzy$.

Figure below illustrates what all of this might look like.



If N is the node corresponding to the occurrence of A farther from the leaf node, the subtree with root node N has height $p + 1$ or less.

It follows that

- $|wxy| \leq 2^{p+1} = n$

The leaf nodes corresponding to the symbols of x are descendants of only one of the two children of N , and because G is in **Chomsky normal form**, the other child also has descendant nodes corresponding to **terminal symbols**. Therefore,

- w and y can't both be Λ (i.e. $|wy| > 0$)

Finally, we have

- $S \Rightarrow^* vAz \Rightarrow^* vwAyz \Rightarrow^* vwxyz$

where the two occurrences of A are the one farther from the leaf node and the one closer to it, respectively. We have already shown how the third conclusion of the theorem follows.

Show that $AnBnCn$ is not a context-free language.

Note: $AnBnCn = \{a^n b^n c^n \mid n \geq 0\}$

Proof: Suppose for the sake of contradiction that $AnBnCn$ is a context-free language, and let n be the integer in the pumping lemma. Let u be the string $a^n b^n c^n$. Then $u \in AnBnCn$, and $|u| \geq n$; therefore, according to the pumping lemma, $u = vwxyz$ for some strings satisfying the **conditions 1–3**.

Condition 1 implies that the string wxy contains at least one symbol, and **condition 2** implies that it can contain no more than two distinct symbols. If σ_1 is one of the three symbols that occurs in wy , and σ_2 is one of the three that doesn't, then the string vw^0xy^0z obtained from u by deleting w and y contains fewer than n occurrences of σ_1 and exactly n occurrences of σ_2 .

- This is a contradiction, because **condition 3** implies that vw^0xy^0z must have equal numbers of all three symbols.
 - **Note:** We could also have obtained a contradiction by considering $m > 1$ in condition 3.

As one can see, we have shown that the string $vxz = vw^0xy^0z$ is not only not an element of $AnBnCn$, but also not an element of the larger language

$$\{t \in \{a, b, c\}^* \mid n_a(t) = n_b(t) = n_c(t)\}$$

and so our argument also works as a proof that this larger language is not a CFL.